

STB 111 FUNGI, PLANT AND ANIMAL TAXONOMY

ANIMAL TAXONOMY BY MR ABDUL ABDUL

WEEK 10

GENERAL CLASSIFICATION OF THE ANIMAL KINGDOM

Members are all multicellular and exist either in water or on land. The animal kingdom is divided into two major groups

Group – invertebrata and

Group – vertebrata.

The invertebrates comprises of animals that do not possess backbone while the vertebrate animals are those that possess backbone

The group invertebrate comprises of the following phyla:

1. Phylum: coelenterata
2. Phylum: Platyhelminthes
3. Phylum: Nematoda
4. Phylum: Annelida
5. Phylum: Arthropoda
6. Phylum: Mollusca

10.1 Group – Craniata (vertebrata)

This forms the major part of the phylum Chordata

General Characteristics of vertebrata

1. Possession of an enlarged brain that is enclosed in a brain case – cranium
2. A segmental spinal column of vertebrae that became the axial support of the body
3. Body comprises of a head, neck, trunk and tail
4. Show a progressive series of structural and functional advances in all organ systems.

Other characteristics include

1. Body covering is a stratified epithelium of epidermis and dermis with many mucous glands in aquatic animals. Most fishes are covered with protective scales. In land dwellers, the exterior is confined with scales on reptiles, feathers on birds and hair on mammals.
2. The internal and jointed skeleton is of cartilage in lower vertebrates and of bone in higher groups. This support and protects various organs
3. Muscles are attached to the skeleton that makes movement possible
4. Complete digestive tract that is ventral to the vertebral column
5. Circulatory system of a well-developed muscular heart of two, three or four chambers
6. Respiration in lower forms is by paired gills while terrestrial forms have lungs
7. Paired excretory organs – kidney. In lower form, it is segmental in nature and drain waste both from the coelom and the blood. In higher forms, it is non segmental and drain only from the blood.

8. Series of endocrine glands that secrete hormones for regulation of body processes.
9. With rare exceptions, sexes are separate

The group vertebrata comprises of the phylum Chordata with the following major classes Pisces (Fish), Amphibia (Amphibians), Reptilia (Reptiles), Aves (Birds) and Mammalia (mammals)

Plas mo de muscles

10.2 Group invertebrata

Phylum Cnidaria

Characteristics of the phylum: Cnidaria or coelenterata

- Tissue grade of organization
- Simple aquatic animals, mostly marine with few fresh water forms e.g. Hydra.
- Radially or biradially symmetrical with a central gastrovascular cavity that opens to the exterior as the mouth i.e. only one opening (mouth with no anus). The mouth is surrounded by tentacles.
- Diploblastic, outer layer called ectoderm and an inner layer of endoderm separated by a non cellular, gelatinous mass called MESOGLEA.
- Fairly small animals but often found joined together to form colonies/units termed ZOOIDS
- Possess nematocyst used for paralyzing prey or holding it.
- No circulatory, respiratory or excretory organs.
- No coelomic cavity
- Nerve system of one or more network of nerve cells but no central nervous system.
- Exhibit polymorphism i.e. two main types of structures, the POLYP and MEDUSA.
- Polyp is sessile and is the asexual form of zooid while medusa is free swimming and is the sexual zooid
- Reproduction is both asexual and sexual. Asexual is by budding and sexual occurs by means of sperm and ova.

- Egg usually develops into a ciliated free swimming larva known as PLANULA e.g. Hydra, obelia; physalia, jellyfish and sea anemone
- This phylum consists of three main classes

Class: Hydrozoa

- Solitary or colonial
- Asexual polyp and sexual medusae present medusae with a velum.
- Mostly marine e.g. obelia, physalia. Freshwater form include Hydra.

Class: Scyphozoa

- Solitary
- Polyp stage reduced or absent
- Bell shaped medusa without a velum
- All are marine e.g. Jellyfish (Aurelia aurita).

Class: Anthozoa

- Solitary or colonial
- Exclusively marine forms
- All polyps, no medusoid stage
- Found attached to stones or seaweed e.g. sea anemone (*Tealia crassicornis*) Corals (*Medusina Cerebriformis*)
- Small solitary polyp living in fresh water size is 10 - 30mm.
- Body is cylindrical tube with lower end to form based disc or foot used for attaching to objects and locomotion. The opposite and free oral end contain mouth on a conical hypostome encircled by 6 - 10 slender tentacles. Mouth leads into a digestive cavity or

gastrovascular cavity or interact. Nematocysts are found throughout the epidermis especially on the tentacles. The side of the body may bear lateral buds that give rise to new individual by asexual reproduction. At times, it bears other rounded projections, the ovaries or testis concerned with sexual reproduction.

10.3 Phylum Platyhelminthes (flat worms)

General characteristics

- Bilaterally symmetrical, non-segmented, hermaphroditic worms.
- Dorsoventrally compressed (flattened), acoelomate (lacks colon), triploblastic animals.
- A system of flame cells and excretory vessels is present.
- No blood vascular system.
- Many are free-living but others are parasitic on or in the bodies of larger animals.
- Single opening to the alimentary canal i.e. the mouth.
- Great power of regeneration.

There are three main classes.

Class: Turbellaria (entirely free living)

Class: Trematoda (entirely parasitic)

Class: cestoda (entirely parasitic)

Example: planaria, fasciola, taenia, Schistosoma

External features of Planaria

These are small creatures with flattened leaf-like bodies. Black in colour, up to 15mm long. Body is broad in front than at the back where it roundly tapered and became extremely flattened.

Has a definite front end, upper dorsal surface and lower ventral surface which is applied to the substratum. Eyes are found near the anterior end. Mouth is at the ventral surface and of the only opening to the gut.

10.4 External features of Fasciola hepatica

Leaf-like structure. Adult has a flattened, oval shaped structure 2.5cm long and 1.2 – 1.3cm wide.

Anterior region has a triangular projection at the apex – main and is surrounded by oral sucker. Ventrally at the base of the projection is a rounded ventral sucker. Between the two suckers is a shallow depression genital atrium. Posteriorly is a minute, terminal excretory pore.

Body is covered by a tough cuticle. Mouth leads into a short buccal cavity followed by muscular pharynx and a narrow oesophagus which fork into two blind branches leading to posterior region and given off numerous lobed caeca.

10.5 Phylum Nematoda (Round Worms)

- Bilaterally symmetrical, triploblastic, unsegmented, pseudo coelomates
- Cylindrical but tapering at the anterior and posterior ends.
- Uniform in structure although differ in size and habitat.
- Can be found in terrestrial, freshwater, marine or parasitic habitats.

- Lack suckers, mouth is surrounded by Lips or papillae
- All are dioecious (separate sexes)
- Alimentary is a straight complete tube with two openings; mouth and anus.
- Species are parasitic in man and vary in size from threadlike object just visible to the naked eye (strongyloides) to elongated string-like worms attaining a length of 60cm (Dracunculus)
- Have internal fertilization,
- Females are either viviparous (produce larva) or oviparous (lay eggs).
- Usually undergo four moults during development from ovum to adult and the third stage larva is responsible for infection of a new host.

- Nematodes which infect human live in the tissues or intestinal tract.
Tissue nematodes are transmitted by insect vector only while intestinal nematodes are soil transmitted.

Nematodes are divided into two main groups based on phasmids (caudal sensory structure)

1. Class: - Aphasmida or Adenophorea
Members lack phasmids. Male has one spicules e.g. *Trichuris trichiura* (whipworm) in soil, *Trichinella spiralis* (found in undercooked or raw meat)
2. Class: Phasmids: Members have phasmids, male has two spicules e.g. *Ancylostoma duodenale* (hookworm), *Dracunculus medinensis* (Guinea worm), *Ascaris Lumbricoides* (roundworm).

Structure of *T. Spiralis*

Small cylindrical worm pointed anteriorly and rounded posteriorly, female is 3mm long, and 80cm wide, male is 1.5mm long and 40cm wide.

Structure of *Ascaris lumbricoides*

- Large worm, body is relatively stout although it tapers at each end particularly at the hinder end.

In male, the tail end curves sharply towards the ventral surface, shows a pair of blister (copulatory spicules) protruding from the cloaca which is situated centimeter away from the tip of the tail.

Three lips guard the entrance of the mouth. Body is encased in a tough but smooth cuticle which is semitransparent so that some of the internal organs are seen through it.

10.6 Phylum Annelida (Ringed Animals)

General Characteristics

- Triploblastic, elongated cylindrical body.
- True (metameric) segmented worms with well-developed nervous system.
- Perivisceral coelom is present.
- Body wall are muscular and have an inner layer or an outer layer of circular muscles
- Excretory system consist of tubes called nephridia
- Body covered by thin cuticle with setae
- Some live in sea, soil or parasitic
- Closed blood system, digestion is extra cellular
- Larva when present is trochophore e.g earthworm, leeches, and nereis.

There are three main classes

Class: Polychaeta

The opisthosoma is divided into a *Mesosoma* with 5 or 6 pairs of lamelli form appendages of which the 1st form a genital operculum while others are biramous and bears gills and a *Mesosoma* without appendages and with a telson. E.g. Horseshoe crab
Subphylum - Crustacean e.g. crabs, shrimps, lobsters, woodlice

Characteristics

Aquatic with majority occurring in marine environment

Exhibit tremendous morphological diversity

Include the largest arthropods, the crabs and lobsters as well as the smallest tiny water fleas and shrimps.

Five of the six head segments bears appendages.

Two pairs of sensory antennae and three pairs of mouth parts.

The remaining segment is divided into two region, thorax and abdomen terminated by a post segment region called telson.

The thoracic segment fused with head to form the cephalothorax while a backward fold of head forms the carapace.

Sexes are separate and often display sexual dimorphism but few are hermaphrodite.

Reproduction is by sexual means but parthenogenesis also occur

Very plentiful in sea therefore termed marine insects.

Subphylum - *Uniramia*

Class - *insecta* e.g. grasshopper, cockroach, ant, fly

General characteristics

Most extensive class in the animal kingdom

The number of described species exceeds the number of all other known animal species
Inhabit all habitable terrestrial environment

Body divided into three regions - head, thorax and abdomen

Head bears six segments, a pair of antenna, three pairs of mouth parts, mandibles, first and second maxillae

Thorax bears three segments and each segment bears a pair of appendages (legs), second and third segments bear wings if present

Abdomen bears eleven segment and posterior telson

Anus opens at the posterior end of the abdomen and just anterior to it are the genital aperture that opens ventrally

Appendage at the eleventh segment is the *Cerci*

Respiration is by direct diffusion of gases

Undergo metamorphosis during development

Depending on feeding habit, mouth part can be modified into sucking, piercing or chewing type.

Class: *Myriapoda*

Characteristics

Body consist of head followed by numerous similar segments

Series of similar legs bearing segment

Single pair of antennal

One or two pairs of maxillae behind mandibles

Mostly terrestrial

Include the centipedes and millipedes

10.9 Phylum: Mollusca

General Characteristics

Second only to arthropods in terms of abundance

- Display a bewildering diversity in form despite similarity in basic morphology

- Range in size from minute of few mm (half mm) snails to giant squids which is over 15m long exclusive of tentacles that is 6m in circumference and weighing 2-3 tonnes. Includes the small flattened chitins which creep about on rocks of sea shore, staggering variety of bivalves, snails, slugs and peculiar burrowing tooth shell of sandy or muddy substratum. The most highly evolved are the predatory cephalopods (Octopus, squids and cuttlefish).

- Wide spread in aquatic and terrestrial habitat.

- Triploblastic

- Coelomate animals which do not show segmentation, soft bodied and bilaterally symmetrical

- Body consist of head, with tentacles flat creeping foot and dome like visceral hump containing most of the alimentary canal, digestive gland, gonads, heart, kidney and paired gills called ctenidia. The skin of the visceral hump is called the mantle which grows downward posteriorly to cover a cavity called mantle cavity into which opens the anus, the excretory and reproductive duct. The mantle secretes the shell.

- Sometimes a larval form which resembles the trochophore larva of annelids

- Of interest to man as food, as source of dyes, intermediate host to many important parasitic helminthes, shells use extensively for ornament, for shell collection

- Mouth is inferiorly while anus is posteriorly located

Six main classes

1. Class: Monoplacophora

2. Class: Amphineura

3. Class: Scaphopoda

4. Class: Gastropoda

5. Class: Bivalvia or Lamellibranchiata

6. Class: Cephalopoda

Class: Monoplacophora - mainly fossil molluscs

Only one living genus neopilina with four known species. Have limpet - like appearance lives in the depth of the ocean. Shell of neopilina is uncoiled. There is a simple foot used for creeping. Gills, kidneys, coelomic sacs and contain muscles are serially repeated.

Class: Amphineura

Constitute a small class of marine mollusks

Represented by several genera and species of chitins, all of which are small, sluggish creatures found in rock crevices or clinging to the undersides of stones.

Possess a dorsal shell made up of longitudinal row of eight plates. The head-foot has a broad creeping sole but the head is inconspicuous and devoid of tentacles or eyes. The mouth is at the anterior end of the head and the anus at the extreme posterior end of the body. They are bilaterally symmetrical.

Class: Scaphopoda

These are small group of burrowing marine mollusks known as the tusk - shells with a hollow that opens at both ends within which is an elongated body with a short extensible foot used mainly for burrowing into the sandy deposits of which the animal live. Ranges in length usually from about 2 - 5cm. The head bears bunches of slender tentacles used for catching food.

The largest living species is *Dentalium vernei* which is about 13cm long and a diameter of 1.5cm.

Class: Gastropoda

Constitutes the largest and most diverse group of the Mollusca with about 40,000 living species and more than 15,000 fossil species. Predominantly marine group, however widely spread in fresh water habitats.

One large sub class pulmonata has become typically terrestrial. Body consist of a ventrally well-developed head and a broad muscular foot both which retain a bilateral symmetry. Exhibit unusual features not found in any other class of mollusk. They are coiling and torsion. Torsion is usually an early larval event, coiling occurs at a later phase and development e.g. periwinkles, topsnails, limpets, sea snails, slugs, land snails e.t.c.

Class: Bivalvia

Mainly marine which burrow in sand and mud but some have colonized the fresh water with no terrestrial forms. It is the second largest class of mollusca. Bilaterally symmetrical, laterally compressed with extensive lateral mantle lobes which secrete a single shell of two valves and a ligament which joins them dorsally at a hinge. Head is rudimentary. Ciliary feeders, the foot is compressed and adapted for burrowing. Fertilization is external

E.g. mussels, scallops, oysters, clams, cockles

Class: Cephalopoda

All are active predaceous creatures able to swim and capture large crustacean such as crabs

- Body is characterized by a fore - shortening of the anterior - posterior axis and an extension of the dorso-ventral axis.
- Head well developed and bears a pair of highly developed eyes and a number of tentacles. Foot is represented by a siphonal apparatus that assist in swimming. Some have ink sac (cuttlefish)
- Sense organs are highly developed. Shell occur with the exception of *Nautilus* where it encloses the body is reduced and internal and in octopuses it is absent. E.g. squid, pearly Nautilus, Octopus, cuttle fish

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PHYLUM CHORDATA

General characteristics

- Largest and ecological most important phylum in line of evolution.
- It is distributed in all habitats- marine, fresh water and terrestrial.
- Include all the large animals present on earth today with the possible exception of the cephalopods.

- Possession of a notochord at least sometimes or another during their bi-cycle. It may persist throughout life e.g. lancelet, lamprey or be replaced either wholly or partially in the adult animal by a backbone example fishes reptiles, birds and mammals
 - Central nervous system is always dorsal in position, hollow and tubular in structure. The anterior end of the cord is usually enlarged to form brain.
 - Bilaterally symmetrical with three germ layers and a segmented body marked externally.
 - Circulation of blood in blood vascular system is maintained by a central muscular propulsive organ-heart, that is ventral (closed circulation) except urochordata.
 - Post and tail usually projecting beyond the anus which is extremely flexible and muscular.
- In aquatic forms, it constitute the main propulsive organ. In terrestrial form, it may be reduced in the adult.

- Complete digestive system.

- Cartilaginous or bony endoskeleton formed in mesoderm.

- Exoskeleton often present, well developed in fishers, reptiles and birds.

- Sexes usually separate (few hermaphrodite or protandric), oviparous or viviparous

The major classes of this phylum which comprises of the vertebrate include:

- Class - Pisces (fishes)

- Class - Amphibia (Amphibians)

- Class - Reptilia (Reptiles)

- Class - Aves (Birds)

- Class - Mammalia (Mammals)

- Class Pisces

General characteristics

- Cold Blooded (poikilothermic)

- Aquatic, either fresh water or marine

- Skin covered with scales

- Body streamlined

- Presence of paired and unpaired fins supported by soft or spiny ray

- Dorsal, caudal and anal fins are unpaired while the pectorals and pelvics are paired

- Endoskeleton by gills, opens to the outside by gill-slits

- Gill-slits are never more than seven pairs usually five pairs are present

- Swim bladder or air bladder usually present

- Lateral line well developed

- Heart is venous and two chambered

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- Class Amphibia

General Characteristics

- Skin moist and glandular, no external scales

- Two pairs of limbs for walking or swimming. No fins, four to five toes or fewer.

- Two nostrils, connected to mouth cavity, eyes often with movable lids, ear-drums, external on toads and frogs. Mouth usually with small teeth, tongue often protrusible.

- Skeleton largely bony, skull with two occipital condyles. Ribs if present not attached to sternum.

- Heart typically three chambered

- Respiration by gills, lungs, skin or mouth lining separately or in combination
- Body temperature variable
- Excretion by mesonephric kidneys
- Fertilization can be external or internal, mostly oviparous, usually an aquatic larva form with metamorphosis to adult form.

Class: Reptilia

General Characteristics

- Body covered with dry cornified skin usually with ecdodermal scales
- Limbs provided usually with five toes and adapted for climbing, running or paddling.
- Absent in some lizards and snakes
- Skeleton well ossified
- Respiration by lungs and gills, cloacal respiration in some aquatic turtles
- Heart incompletely 4 chambered
- Body temperature variable
- Excretion by metanephric kidneys
- Sexes separate, fertilization internal
- Amniotic eggs covered with calcareous or leathery shells e.g. lizard, crocodiles, Snakes and turtle

Class - Aves (birds)

Characteristics

- Body spindle shaped with four division - head, neck, trunk and tail
 - Body covered with feathers
 - Two pairs of limbs, forelimbs modified as wings for flight, hind limbs adapted for perching, walking or swimming (with lobes or webs)
 - Skeleton light, strong, fully ossified with air sacs. Many bones fused, providing rigidity
 - Upper and lower jaws drawn in front to form beak or bill with horny sheath. No teeth.
 - Skull with are occipital condyle
 - Neck usually long and flexible
 - Respiration by compact highly efficient lungs attached to ribs and connected to thin walled air sac
 - Voice boxes at base of trachea.
 - Body temperature constant.
 - Excretion by metanephric kidneys. Fertilization internal, female with only left ovary and oviduct.
 - Young at hatching cared by parent
- #### Class Mammalia
- Body covered with hair but reduced in some. Skin with many glands (Sebaceous, sweat, scent and mammary)
 - Skeletal features of the skull with two occipital condyles, seven cervical vertebrae and often an elongated tail
 - Mouth with teeth on both jaws
 - Movable eyelids and fleshy external ears
 - Fore limbs (reduced or absent in some) adapted for many forms of locomotion

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- Four chambered heart, persistent left aorta
 - Respiratory system of lungs and voice box.
 - Muscular partition (DIAPHRAGM) between thorax and abdomen
 - Excretory system of metanephric kidneys with ureters that opens into a bladder
 - Nervous system of well-developed brain and twelve pairs of cranial nerves
 - Sexes separate with reproductive organs e.g. vagina.
 - Internal fertilization. Eggs develop in a uterus with placental attachment.
 - Young nourished by milk from mammary glands with exception of Duckbilled platypus and spiny anteaters that lay eggs and do not possess true mammary glands.

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EXTERNAL FEATURES OF SOME CHORDATES

External Features of Tilapia

It is found in an aquatic environment (fresh water). Bony fish, spindle shaped, flattened laterally to give it a streamlined shape that allows it to move smoothly through the water. Body is covered with scales with paired and unpaired fins. Unpaired fins are the dorsal, ventral and tail fins while the pairs of fins are the pectoral and pelvic fins.

External Features of Frog

The skin of frog is soft, smooth and moist. The head bears a wide mouth, small valvular nostrils, large spherical eyes and behind each eye are flat eardrum or tympanic membrane. Each eye has a fleshy, opaque upper eye lid and a lesser lower lid. Continuous with the latter and folded behind it when at rest is a transparent portion which functionally resembles the nictitating membrane of higher vertebrates. At the posterior end of the body is the vent or cloacae opening through which are discharged undigested food wastes, urine and eggs or sperm.

At mid body is the sacral hump, where the pelvic girdle is hinged to the sacrum.

The short front leg (arm) comprises of an upper arm, fore arm, wrist and hand with 4 digits and vestigial thumb.

The inner digit (thumb) is thickened on males especially in the breeding season.

The hind leg includes a thigh, lower leg, ankle and long foot with a narrow sole and five slender toes connected by broad thin webs. The palm and soles bear cornified tubercles that protect the feet against abrasion and provide traction on wet, slippery surfaces.

External Features of a Lizard

Body is made up of a head and trunk joined together by a definite neck. The trunk ends in a long tapering tail.

On the head, there is a wide terminal mouth, a pair of nostrils, a pair of slightly protruding eyes with movable eye lids and a pair of ear drums found in shallow depression on the skin. Below the eardrum is a fold of skin, the gular fold which is lowered during courtship or when the animal is frightened. Above the neck is a median crest of skin, the nuchal crest, which the male raises when fighting with each other.

The trunk bears a pair of fore and hind limbs which terminate each in five clawed digits on the underside. Just behind the hind limbs is a transverse slit, the cloaca. In the male, there is a hemi cod at each end of the cloaca.

- Largest class of annelids
- Mostly marine with few found in brackish water, fresh water and moist soil.
- Some are free living, some tube living and some burrows into mud.
- Have well marked internal and external segmentation.
- Each body segment bears a pair of appendages called parapodia which bears numerous chaetae
- Many reach up to a length of 1m.
- Cephalisation is well pronounced.
- Well defined head region with eye and tentacles
- Parapodium used for locomotion assisted by chaetae
- Sexes separate No permanent gonads
- Fertilisation is external with a free swimming trochophore larva. Examples are Nereis diversicolor (rag worm), Arenicola (lugworm), Chaetopterus (tube living worm)
- Class: Oligochaeta
 - Terrestrial or fresh water habitat.
 - Well marked external and internal segmentation
 - Posses few chaetae e. No parapodium
 - Hermaphrodite, cross fertilization
 - Clitellum present at sexual maturity
 - Eggs are deposited in cocoon.
 - No free larval stage
 - Respiration over the entire body surface example earthworm, tubifex
- Class: Hirudinea
 - Vermiform, dorsoventrally flattened
 - Body consist of definite number of segment marked externally by secondary rings and annuli
 - Capable of elongation and contraction to different proportion
 - Feed by sucking in organic matter using pharynx or may be parasitic by sucking blood of mammals, tregadoles and small fishes
 - Found in ponds, swamps and slow flowing streams
 - Hermaphrodite
 - Lack setae and parapodium. Example *Hirudo medicinalis*

10.7 phylum: Arthropoda (Jointed limbs)

In terms of number of species and individuals, this phylum surpass any other phylum. Of some 1 ¼ million known animal species, 4/5 are arthropods. Also this phylum contain the class insecta which represent almost ¾ of described animal species.

General Characteristics of this Phylum

- Triploblastic, bilaterally symmetrical and metamerically segmented
- Paired jointed appendages on each metamer with at least one pair acting as a set of jaws.

Reduced coelomic cavities and a major body cavity – *Haemocoel* with fluid called *Haemolymph*.

Presence of a skeletal covering composed of chitin. This forms the exoskeleton.

- Nervous system with cerebral ganglionic mass connected to ventral paired ganglia
- Open circulation with no blood vessels
- Anterior photoreceptors
- Molt and metamorphosed between immature and adult stage during growth
- Separate sexes
- Can be found almost in every environment including aerial one

Classification

There is no definite system of classification for the phylum existing in view of divergent ideas concerning the phylogeny of arthropods. It is in a state of flux and may always remain.

Divided into four subphyla

1. Subphylum – *Trilobita* (extinct)
2. Subphylum – *Chelicerata* consisting of the class: Arachnida, Merostomata and Pycnogonida
3. Subphylum – Crustacean
4. Subphylum – Uniramia consisting of the classes

Insecta

Myriapoda

Chilopoda

Symphyla

Diplopoda

Pauropoda

*Subphylum – *Chelicerata*

Absence of antenna, presence of a pair of chelate appendages as the first pair of segmental appendages.

Class: *Arachnida* e.g. – spiders, scorpions, Opiliones (harvestman or daddy longlegs)

General Characteristics

Body divided into two parts, bearing six pairs. First pairs- anterior prosoma (cephalothorax) which bears the chelicerae that is associated with the mouth (pre oral), second pairs bears the pedipalps (post oral). The remaining four pairs are legs. Posterior opisthosoma makes up the abdomen. The genital opening is on the anterior part of the opisthosoma. The only appendages here are the pectines of scorpions and spinnerets of spiders

Respiration is by lung books and or tracheae.

Development is straight forward without metamorphosis

Generally terrestrial

Class – *Merostomata*

Characteristics

Aquatic, usually marine with brachial respiration

Body divided into two – prosoma and opisthosoma

Prosoma has chelicerae on the first segment and five similar pairs of appendages on the 2nd 6th segments.